Andrew Miller

Backend Software Engineer · Kotlin · Fintech

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Backend engineer with four years on the risk platform at EarnIn, an earned-wage-access fintech, building the Kotlin and gRPC services that set cash-out limits and repayment timing for members. Drove the move off legacy queue processors onto a shared service and retired seven deployables. Language-agnostic (Kotlin, Java, Python, C#) and proficient in AI-assisted development with Cursor and Claude Code.

SKILLS

Languages: Kotlin, Java, Python, C#, JavaScript, SQL, Groovy, C++
Backend: gRPC, Protobuf, Ktor, Kotlin Coroutines, kotlinx.serialization, Spring, Vert.x, .NET Core, REST APIs
Data and messaging: Kafka (Confluent), AWS EventBridge, SQS, SNS, MySQL, MongoDB, Redis, Databricks
Cloud and CI/CD: AWS (Lambda, S3), Kubernetes, Docker, Terraform, Gradle, GitHub Actions, Jenkins, Spinnaker
Testing and observability: JUnit 5, Testcontainers, xUnit, ktlint, detekt, Datadog (APM, logs, monitors, SLOs)
AI and tools: Cursor, Claude Code, feature flags (OpenFeature, Optimizely), Retool, React

EXPERIENCE

Software Engineer — June 2022 – October 2026

EarnIn (Activehours, Inc) · Mountain View, California

- Designed and launched a shared Kotlin/gRPC risk-profile service fed by EventBridge, SQS, and a Python Lambda, from the inception doc and phased rollout plan through production monitoring.
- Drove the 2026 infrastructure-simplification plan: wrote its inception doc and runbook, moved six legacy queue processors to shared services via flag-gated ramps (1% → 25% → 100%), retiring seven deployables.
- Retired a legacy risk service by moving its last production callers (payroll and customer-care systems) onto Kotlin gRPC endpoints with feature-flag ramps and shadow comparison.
- Built the backend for an experiment that raises cash-out limits for members on a paycheck-deposit streak: new policy facts and limit metadata, seven policy updates with data science, and policy macros.
- Shipped limit features across products: a real-time checkpoint for an ML early-pay limit model, a card-activation boost made permanent after positive results, and repayment handling for EarnIn Payroll.
- Launched risk-policy execution on an open-source rules engine and mitigated unpatched CVE-2023-50571 with an unsafe-rule filter. Built the gRPC backend and Retool admin for fraud policies, with two-person approval.
- On-call first responder for risk services: root-caused a bug that skipped restore-date events for about 19% of eligible members (~2,960 users), fixed duplicate event dispatch from an undersized SQS visibility timeout, and traced 12,000+ gRPC SLO errors to downstream incidents.
- Built per-queue Datadog dashboards, and removed payload logging from two Lambda proxies, cutting their projected log volume 40–64% (~6.8M fewer logs a week) and keeping member identifiers out of logs.
- Built an AI review step that flags Terraform changes that would destroy infrastructure other services depend on (incident follow-up), a shared Testcontainers test framework, and a fraud-policy regression simulator.
- Security champion: security reviews, design review and threat model for a third-party fraud platform, and fixes for 19 high and critical vulnerabilities. Organized onboarding for new engineers.

Software Developer I · Battelle Memorial Institute · Columbus, Ohio — May 2021 – May 2022

- Built full-stack search for a public CDC Data Explorer; worked directly with CDC stakeholders.

Software Developer Co-op · Myriad Neuroscience · Mason, Ohio — January 2020 – August 2020

- Raised unit-test coverage to cut QA load, and added component tests to the microservice template.

Web Developer Co-op · Fortech LLC · Cincinnati, Ohio — Aug 2018 – Dec 2018 · May 2019 – Aug 2019

- Built .NET Core MVC and React apps, including an MES dashboard for P&G plant-floor staff.

PROJECTS

- **api-sports-kotlin** (Maven Central): Kotlin client libraries for the NBA and NFL sports-data APIs on a shared core (Ktor, coroutines, kotlinx.serialization, KSP), with GitHub Actions CI, Dokka docs, and automated releases.
- **Discord bots** (Kotlin, C#): gRPC and Protobuf microservices with MongoDB, Docker, and GitHub Actions CI.

EDUCATION

University of Cincinnati — August 2016 – May 2021

B.S. Computer Engineering · Cincinnati, Ohio · Cincinnatus Scholar · Eagle Scout